Practice 4: Heart Disease Classification using Multilayer Perceptron

The goal of this practice is to classify patients from the UCI Heart Disease dataset into the categories of “Healthy” (0) and “Sick” (1, 2, 3, or 4). The data was binarized before proceeding.

Artificial Neural Networks are sensitive to the scale of inputs, therefore all data was standardized using StandardScaler. I chose 5-Fold Cross-Validation since it’s ideal for a dataset of this size (approximately 300 samples) because it ensures every patient record is used for testing,and thus providing an estimate of the model’s performance we can trust more compared to other methods.

Experiment 1: Full Feature Set

Here, all 13 original features were used to train the Multilayer Perceptron. Testing 3 activation functions (logistic, tanh, relu) and three hidden layer architectures: (50,), (100,), and (50, 50).

Activation	Hidden Layers	Accuracy	F1-Score
logistic	(50,)	0.8250	0.8002
logistic	(100,)	0.8316	0.8089
logistic	(50, 50)	0.8316	0.8092
tanh	(100,)	0.8413	0.8201

tanh	(50, 50)	0.8248	0.8014
relu	(50,)	0.8115	0.7903

For this experiment the best performance was achieved when using the tanh activation function with a 1 hidden layer of 100 neurons, reaching an accuracy of 84.13%. The logistic function performed well, while relu showed a bit lower results here. Implying that for small datasets like this one, other activation functions like tanh can be better at detecting stuff that relu sometimes can't.

Experiment 2: Reduced Feature Set (5 Features)

In this case only 5 features were taken into account: age, sex, chest pain, blood pressure, and cholesterol.

Activation	Hidden Layers	Accuracy	F1-Score
logistic	(50, 50)	0.7757	0.7590
relu	(50,)	0.7657	0.7412
tanh	(100,)	0.7624	0.7395
relu	(50, 50)	0.7161	0.6874

Reducing the features to just 5 caused a drop in performance, with the highest accuracy dropping to 77.57%. Here even the most complex model of (50, 50) had a hard time compared to Experiment 1. This

was expected as the model loses access to variables that help predict better, like max heart rate or oldpeak. However, 77% accuracy using only basic clinical data is quite solid, and comes to show that a Multilayer Perceptron can still extract significant patterns from a limited feature space.

Conclusions

This practice highlights two main takeaways. First, that Hyperparameter Tuning is very important; there isn't a network that works the best for everything, as we saw with tanh doing better than relu. Second, no amount of extra hidden layers can compensate for the loss of information when we moved from 13 to 5 variables. A simpler model may be easier to implement and use less resources, but regarding the medical context and patient safety, high precision is required.

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Experiment 1
| Activation    | Hidden Layers    |   Accuracy |   F1-Score |
|:-------------|:----------------|----------:|----------:|
| logistic      | (50,)            |   0.824973 |   0.800205 |
| logistic      | (100,)           |   0.831639 |   0.808918 |
| logistic      | (50, 50)         |   0.831639 |   0.809205 |
| tanh          | (50,)            |   0.834809 |   0.813337 |
| tanh          | (100,)           |   0.841311 |   0.82009  |
| tanh          | (50, 50)         |   0.824809 |   0.80138  |
| relu          | (50,)            |   0.81153  |   0.790348 |
| relu          | (100,)           |   0.801694 |   0.779986 |
| relu          | (50, 50)         |   0.801475 |   0.771725 |
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Experiment 2
| Activation    | Hidden Layers    |   Accuracy |   F1-Score |
|:-------------|:----------------|----------:|----------:|
| logistic      | (50,)            |   0.762404 |   0.739283 |
| logistic      | (100,)           |   0.762404 |   0.745123 |
| logistic      | (50, 50)         |   0.775683 |   0.75903  |
| tanh          | (50,)            |   0.752568 |   0.725667 |
| tanh          | (100,)           |   0.76235  |   0.739492 |
| tanh          | (50, 50)         |   0.759071 |   0.740901 |
| relu          | (50,)            |   0.765738 |   0.741173 |
| relu          | (100,)           |   0.75918  |   0.733486 |
| relu          | (50, 50)         |   0.716066 |   0.687395 |

Process finished with exit code 0
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